MIL-P-10398C
13 December 1982
SUPERSEDING
MIL-P-10398B
11 February 1969

MILITARY SPECIFICATION

PHOSPHORUS TRICHLORIDE, TECHNICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one technical grade of phosphorus trichloride.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1188	-	Ammunition Lot Numbering

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6810

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS (CFR)

40 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFR 171 to 179.")

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM STANDARDS

D86 - Distillation of Petroleum Products
D891 - Specific Gravity, Apparent, Of Liquid Industrial Chemicals
D1193 - Reagent Water
D1209 - Color of Clear Liquids (Platinum - Cobalt Scale)

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Physical characteristics. Phosphorus trichloride shall conform to the physical characteristics of table I when tested as specified therein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the

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TABLE I. Physical characteristics

Characteristic	Minimum	Maximum	Test paragraph
Color, platinum-cobalt scale	----	No. 5	4.2.4.1
Specific gravity, 15.5°/15.5°C	1.582	1.588	4.2.4.2
Distillation range (°C):			4.2.4.3
Initial boiling point	74.5	----	
Dry point	----	79.0	
Total range	----	3.0	

right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the phosphorus trichloride produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3). Each lot shall be identified and controlled in accordance with MIL-STD-1168.

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test (see 6.5). Sampling for test shall be conducted in accordance with table II. A representative specimen of approximately 500 milliliters shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for test

Number of containers in batch or lot	Number of sample containers
3 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

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4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled shipping container, ready for shipment. Sample shipping containers shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Marking incorrect, missing, or illegible

4.2.3.2 For test (see 6.5). Nitrogen blanketing shall be used when transferring specimens to avoid reaction with air. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests (see 6.5). Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Specimen transfer shall be accomplished under a nitrogen blanket. All apparatus shall be purged with nitrogen, and a nitrogen purge shall be maintained where applicable. Tests shall be conducted as follows:

4.2.4.1 Color. Determine the color of the specimen in accordance with ASTM D1209.

4.2.4.2 Specific gravity. Determine the specific gravity of the specimen in accordance with ASTM D891.

4.2.4.3 Distillation range. Determine the initial boiling point, dry point, and total distillation range of the specimen in accordance with ASTM D86, correcting to 760 millimeters of mercury.

5. PACKAGING

5.1 Unit packing, level C. A specified quantity of phosphorus trichloride (see 6.2) shall be packed in accordance with Department of Transportation (DOT) regulations. Threads on the plugs of any drums shall be luted with graphite grease.

5.2 Packing, level C. Phosphorus trichloride, unit packed as specified in 5.1, shall be packed in accordance with DOT regulations if further protection is required for shipment.

5.3 Marking. Shipping containers of phosphorus trichloride shall be marked in accordance with DOT regulations and MIL-STD-129. The "Corrosive" white label specified in DOT regulations shall be applied to all shipping containers. In addition, each container shall bear the following marking:

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HAZARD:

Corrosive, Eye and Skin.
DANGER! CAUSES SEVERE BURNS.
Do not get in eyes, on skin, on clothing.
Contact lenses should not be worn when working with this chemical.
Do not breath vapor, extremely irritating.
Do not open container unless you have been properly instructed.
Use with adequate ventilation.
Do not add water to contents while in a container because of violent reaction and possible flash fire.
Do not clean or re-use this container.
In case of spillage: provide adequate ventilation; collect for reclamation or absorb in vermiculite, dry sand, earth, or similar material; or cover with sodium bicarbonate or an equal mixture of soda ash and slaked lime, dilute carefully with a small spray of water, then excess water, and dispose in a secured sanitary landfill.

FIRST AID: In case of contact, immediately remove all contaminated clothing and shoes. Flush affected areas with plenty of water for at least 15 minutes. Call for medical aid. Wash clothing before re-use. (Discard contaminated shoes).
IF IN EYES, hold eyelids open and flush with plenty of water.
IF SWALLOWED and victim is CONSCIOUS have victim drink water or milk.
DO NOT INDUCE VOMITING.
If breathing has stopped, give artificial respiration. If breathing is difficult, give oxygen.

6. NOTES

6.1 Intended use. Phosphorus trichloride is intended for use as a chlorinating agent and as a phosphorus solvent.

6.2 Ordering data. Acquisition documents should specify the title, number, and date of this specification and the unit quantity of phosphorus trichloride required (see 5.1).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

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6.5 Sampling and testing precautions. This specification covers inspection of chemical material which is potentially dangerous to personnel. Phosphorus trichloride is corrosive, harmful to eyes and skin. All applicable safety rules, regulations, and procedures must be followed in the handling and processing of this material.

6.5.1 Safety. Phosphorus trichloride should be handled with care as its fumes are poisonous and flammable. Contact is extremely hazardous and may cause serious burns to eyes and skin. Vapors are extremely irritating. Phosphorus trichloride decomposes rapidly upon contact with atmospheric moisture. Contact with water may cause flash fire.

6.5.2 Sampling. Personnel drawing specimens should wear impervious gloves and clothing in addition to splashproof safety goggles and full face shield. Specimen containers should be glass bottles with snap-top polyethylene screw caps. A 1/2-inch diameter hole should be punched in the tip of the cap and a 1/16-inch "Koroseal" diaphragm with a 3/8-inch slit should be placed between the cap and the bottle neck. The specimen bottle should be purged with nitrogen prior to use. To sample, attach a 1/4-inch outside diameter sampling tube equipped with a sampling valve to the container being sampled. Fill the specimen bottle by forcing it up over the 1/4-inch sampling tube and opening the sampling valve. When the specimen bottle is approximately two-thirds full, close the valve and remove the specimen bottle from the sampling tube. The slit in the diaphragm will remain closed when not forcibly opened, thereby excluding air and moisture before and after sampling. Further protection is achieved when the snap-cap is replaced after the filling operation. After use, the polyethylene cap should be removed and the specimen bottle diaphragm should be discarded. Wash the cap with water and dry at room temperature before re-use. Caps which show contamination or damage should be replaced.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 6810-B351

Review activities:

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

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c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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